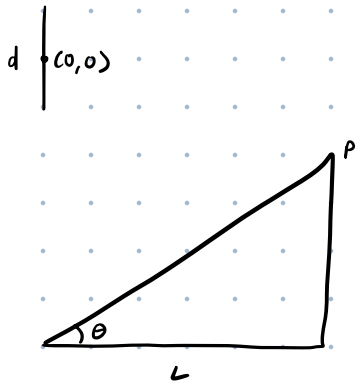
• $P = (x, L\tan\theta)$



$$\vec{r}_1 = L\hat{z} + (L\tan\theta - d/2)\hat{y}$$
$$\vec{r}_2 = L\hat{z} + (L\tan\theta + d/2)\hat{y}$$

$$|r_1| = \sqrt{L^2 + (L\tan\theta - d/2)^2} = L\sqrt{\frac{1}{\cos^2\theta} - \frac{d}{L}\tan\theta + \frac{d^2}{4L^2}}$$

$$|r_2| = \sqrt{L^2 + (L\tan\theta + d/2)^2} = L\sqrt{\frac{1}{\cos^2\theta} + \frac{d}{L}\tan\theta + \frac{d^2}{4L^2}}$$

#2

$$|r_2 - r_1| = L\sqrt{\frac{1}{\cos^2\theta} + \frac{d}{L}\tan\theta} - L\sqrt{\frac{1}{\cos^2\theta} - \frac{d}{L}\tan\theta} \quad \text{to first order}$$

$$\frac{1}{\cos\theta}\sqrt{1 + \frac{d}{L}\sin\theta\cos\theta} - \frac{1}{\cos\theta}\sqrt{1 - \frac{d}{L}\sin\theta\cos\theta}$$

$$|r_2 - r_1| = \cancel{L\frac{1}{\cos\theta}}\left(\cancel{1} + \frac{d}{2L}\sin\theta\right) - \cancel{L\frac{1}{\cos\theta}}\left(\cancel{1} - \frac{d}{2L}\sin\theta\right.$$

$$\boxed{= \frac{d}{L}\sin\theta}$$ #3

$$E = \frac{A}{r_1}e^{ikr_1} + \frac{A}{r_2}e^{ikr_2}$$

$$\boxed{I = |E|^2}$$ #4

$$= EE^*$$

$$= \left(\frac{A}{r_1}e^{ikr_1}\right)^2 + \left(\frac{A}{r_2}e^{ikr_2}\right)^2 + \underbrace{\left(\frac{A}{r_1}e^{ikr_1}\right)\left(\frac{A^*}{r_2}e^{-ikr_2}\right)}_{\frac{A^2}{r_1r_2}e^{ik(r_1-r_2)}} + \frac{A}{r_1r_2}e^{ik(r_2-r_1)}$$

$$= \frac{A^2}{r_1^2}e^{2ikr_1} + \frac{A^2}{r_2^2}e^{2ikr_2} + \frac{A^2}{r_1r_2}e^{ik(r_1-r_2)} + \frac{A^2}{r_1r_2}e^{ik(r_2-r_1)}$$